

Importer les packages python suivants :

```
import numpy as np
from scipy.optimize import linprog
import math
import matplotlib.pyplot as plt
```

Saisir et expliquer le résultat des commandes suivantes :

```
> A = np.array([[1,2],[3,4]])
> A[0,0]
> A[:,1]
> A[0,:]
> n1,nc=A.shape
> print(n1,nc)
> c=np.array([[4,-2,-2]])
> print(type(c))
> print(c.ndim)
> m=np.amax(c)
> print(m)
> print(np.argmax(c))
> n=np.amin(c)
> print(np.argmin(c))
> a = np.array([[1.2,2.5],[3.2,1.8],[1.1,4.3]])
> v = np.arange(0,10).reshape(2,5)
> print(a.reshape(3,2))
> print(np.zeros(shape=(2,4)))
> b = np.array([[4.1,2.6]])
> c = np.append(a,b,axis=0)
> print(c)
> d = np.array([[7.8],[6.1],[5.4]])
> print(np.append(a,d,axis=1))
> print(np.insert(a,1,b,axis=0))
> print(np.delete(a,1,axis=0))
> v = np.array([[1.2,2.5],[3.2,1.8],[1.1,4.3]])
> print(v[0:2,:])
> print(v[-1,:])
> print(v[-2,:])
> print(np.max(v,axis=0))
> s = 0
> for i in range(0,v.shape[0]):
    for j in range(0,v.shape[1]):
        print(v[i,j])
        s = s + v[i,j]
> print("Somme = ",s)
> def racine(x):
    return math.sqrt(x)
> print(racine(4))
> l=list()
> l.append(2)
> print(l)
> x = np.linspace(0, 10, 30)
> y = np.sin(x)
> plt.scatter(x, y, marker='o')
```

```
> rng = np.random.RandomState(0)
> x = rng.randn(100)
> y = rng.randn(100)
> colors = rng.rand(100)
> sizes = 1000 * rng.rand(100)
> plt.scatter(x, y, c=colors, s=sizes, alpha=0.3, cmap='viridis')
> plt.colorbar()
```